

Section
Registers
Reset Values
Register Accesses
Bit Level Register Accesses
APB Protocol
Transmitter
Character Formats
TX FIFO Empty Flag
TX Empty Flag
Receiver
Character Formats
Data Received Flag
RX Line Status
Framing Error
Parity Error
Break Indication
Overrun Error
FIFOE
Status
Modem Interface
Modem Outputs
Modem Inputs
Loopback Mode
Interrupts
Interrupt Enable
Interrupt ID
Receive FIFO Interrupt
Receive Line Status Interrupt
Transmit Empty Interrupt
Modem Status Interrupt
Receive Timeout Interrupt
Baud Rate
Divider Values
Code Coverage
Statement Coverage
Branch Coverage
FSM Coverage

Description

All registers return the specified reset values

All registers have been accesses for all possible access modes

All read-write bits in the registers toggle correctly

APB Protocol has been tested in all modes

All possible character formats are transmitted correctly

The FIFO empty flag is set when the FIFO is empty and is read back correctly

The transmit empty flag is set correctly and is read back correctly

All possible character formats are received correctly

The data received flag is set when data is available and is read back correctly

Framing errors are detected for one or two stop bits

Parity errors are detected for all types of parity

A break condition is detected correctly for all character formats

RX overrun is detected for all character formats

That the FIFO error condition is valid for all error/indication types

Any valid combination of error/indicator has been observed

All combinations of modem output values have been seen

All combinations of modem input values have been seen

The modem input status change signals work correctly

Modem output bits are routed to the right modem status bits

All combinations of the interrupt enable bits have been used

All valid interrupt IDs have been detected

Seen for all possible character formats

All possible RX FIFO Threshold values checked

Interrupts generated for all possible combinations of errors and indicators for all character formats

Generated for all character formats

Generated for all combinations of the signal change bits

Has been checked for the shortest and longest character format and 4 other formats

Check UART operation for a range of baud rate divider values

Check baud rate divider ratio for a selection of values via baud rate divider output

Check each executable line of the RTL has been covered

Check each branch in the RTL has been taken

Each arc in the RTL FSMs has been taken

Coverage Type	Priority
Test Result	1
Covergroup, Cross	1
Test Result	1
APB Monitor	1
Covergroup, Cross	1
Design Assertion, Covergroup	1
Design Assertion, Covergroup	1
Covergroup, Cross	1
Design Assertion, Covergroup	1
Design Assertion, Covergroup	2
Design Assertion, Covergroup	1
Covergroup, Cross	2
Covergroup, Cross	2
Covergroup, Cross	2
Covergroup, Cross	2
Covergroup, Cross	3
Covergroup, Cross	3
Design Assertion, Covergroup	3
Covergroup	2
Covergroup, Cross	1
Covergroup, Cross	1
Covergroup, Cross	1
Covergroup, Cross	2
Covergroup, Cross	1
Covergroup, Cross	1
Covergroup, Cross	3
Covergroup	4
Covergroup	1
Covergroup	2
	1
	1
	1